

**B.Tech. Degree III Semester Regular/Supplementary Examination
November 2023**

**CS 19-202-0306 DATA AND COMPUTER COMMUNICATION
(2019 Scheme)**

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Explain and calculate digital transmission over different types of communication media.
 CO2: Describe the principles of access control to shared media and carry out performance calculations.
 CO3: Solve issues in networking by referring to problem solving steps through relevant information by choosing suitable techniques.
 CO4: Explain the role of protocols in networking.
 CO5: Analyse the services and features of various communication devices.
 Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create
 PO – Programme Outcome

PART A(Answer *ALL* questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
1.	(a)	Discuss the five components of data communication systems.	3	L2	1	1,2
	(b)	What is an IP address and port number?	3	L2	1	1,2
	(c)	What does the Shannon capacity have to do with communications?	3	L1	1	1,2
	(d)	Explain wavelength division multiplexing.	3	L1	2	1,2
	(e)	Discuss hamming distance with suitable example.	3	L2	2	1,2,3
	(f)	Discuss the different methods in error correction.	3	L1	3	1,2
	(g)	Discuss run length encoding with example.	3	L2	3	1,2
	(h)	Discuss about any connecting device that operate at physical layer.	3	L2	4	1,2,3

PART B

(4 × 12 = 48)

II.	(a)	Compare the characteristics of different transmission modes.	6	L1	1	1,2
	(b)	Discuss bit rate and baud rate. If the bit rate of a signal is 3000 bps and each signal unit carries 6 bits, find the baud rate.	4	L3	1	1,2,3
OR						
III.	(a)	Write a short note on delta modulation.	7	L1	1	1,2
	(b)	What is on-off keying?	5	L2	1	1,2
IV.		List several transmission media for networking. Explain each in detail.	12	L1	2	1,2
OR						
V.		Explain the concept of Time division multiplexing. Differentiate between statistical TDM and Synchronous TDM using suitable diagrams.	12	L3	2	1,2,3

(P.T.O)

BTS-III(R/S)-11-23-2170

	Marks	BL	CO	PO
	6	L3	3	1,2,3
VI. (a) Generate the Huffman code and find the average code length with five possible symbols (A, B, C, D, E). The probabilities are $P(A) = 0.11$, $P(B) = 0.4$, $P(C) = 0.16$, $P(D) = 0.09$, $P(E) = 0.24$.				
(b) Given the dataword 10011101 and the generator polynomial is $x^3 + 1$, show the generation of the CRC codeword at the sender site using binary division.	6	L3	3	1,2,3
OR				
VII. (a) Explain go back n ARQ with suitable diagram.	8	L2	3	1,2,3
(b) A 20 Kbps satellite link has a propagation delay of 400 ms. The transmitter employs the "go back n ARQ" scheme with n set to 10. Assuming that each frame is 100 bytes long, what is the maximum data rate possible?	4	L3	3	1,2,3
VIII. Identify the working principle of Bluetooth and develop a neat sketch to depict its architecture.	12	L3	4	1,2
OR				
IX. Explain different types of network topologies. Write each of its advantages and disadvantages.	12	L2	4	1,2

Bloom's Taxonomy Levels

L1 = 42%, L2 = 31.5%, L3 = 26.5%.
